Bellman Defect Correction

Inference Time Algorithms for SciML and GenAI

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More thought can improve the answer

• Flash

Quick response

YOU

A bat and a ball cost $1.10 together.
The bat costs $1.00 more than the ball.
What does the ball cost?

AI

The ball costs **$0.10**.
$1.10 - 1.00 = 0.10$

✗ Fails the check: $0.10 + $1.10 = $1.20

• Pro

Reason and verify

YOU

A bat and a ball cost $1.10 together.
The bat costs $1.00 more than the ball.
What does the ball cost?

AI

Let the ball cost x. The bat then costs $x + 1$.

$$x + (x + 1) = 1.10 \implies x = 0.05$$

The ball costs **$0.05**.

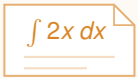
✓ Check: $0.05 + $1.05 = $1.10

Same question. More computation for reasoning and checks.

Illustrative AI dialogues

From fast answers to scientific discovery

Flash mode



Solve undergraduate problems

Reasoning



Develop and check research ideas

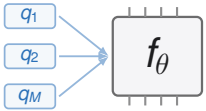
Large agent teams



Navier–Stokes blowup construction*

OUR SCALING SETUP

More queries.
Same model.



Fixed weights θ

More inference computation

Which algorithm turns more queries into better answers?

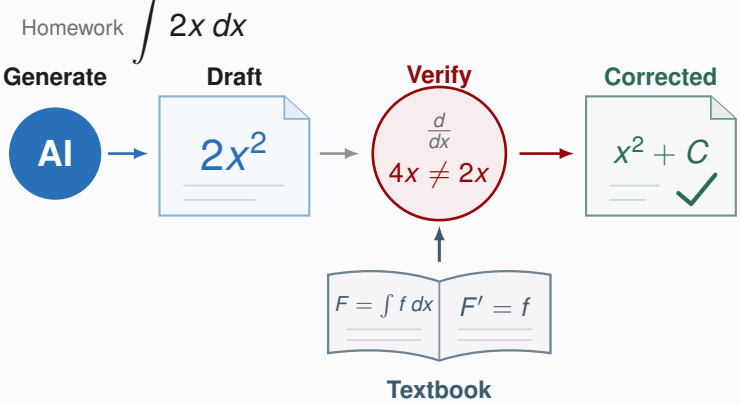
Illustrative task progression. * Reported NS case: ∼10,000 agents, with model upgrades. OpenAI (2026).

Textbook knowledge checks an AI answer

MY AI PHILOSOPHY

Generate first. Verify with known rules.

The error tells us what to correct.



Defect correction has exactly two steps

Write the Law of Defect

Derive the equation obeyed by $\check{u} := u - \widehat{u}$.

1

known equation + autodiff

Simulate the Law of Defect

Spend new compute on stochastic paths for $\check{u}$.

2

linear: Feynman Kac + Monte Carlo

semilinear: multilevel Picard

$$u_{\mathrm{corr}} = \widehat{u} + \widehat{\check{u}}_N$$

Write analytically. Spend inference compute stochastically.

No retraining: additional compute targets only what the model missed.

Source: SCaSML paper, structure preserving Law of Defect and stochastic correction.

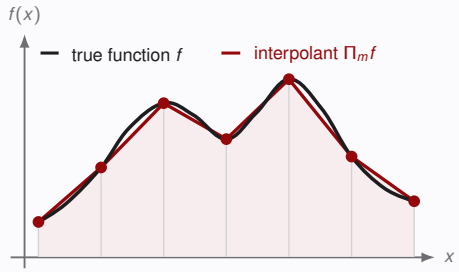
NUMERICAL ANALYSIS 101

Integration from random samples

Rare events, minimax rates, and optimal correction.

Linear case: function approximation

For intuition, take $q = 1$ and $I(f) = \int_\Omega f \, d\mu$.



deterministic nodes reconstruct the integrand

APPROXIMATE, THEN INTEGRATE

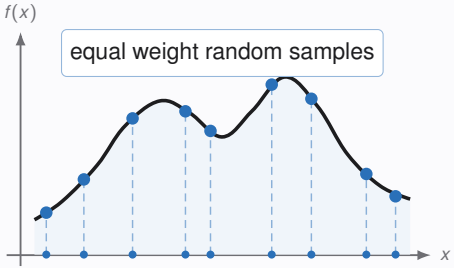
$$f \approx \Pi_m f = \sum_{j=1}^{m} f(x_j)\ell_j,$$

$$\boxed{Q_m(f) = \int_\Omega \Pi_m f \, d\mu = \sum_{j=1}^{m} w_j f(x_j).}$$

APPROXIMATION RATE

$$\begin{aligned} |I(f) - Q_m(f)| &\le \|f - \Pi_m f\|_{L^1(\mu)} \\ &\lesssim m^{-s/d}. \end{aligned}$$

Linear case: Monte Carlo



random nodes estimate the area directly

SAMPLE, THEN AVERAGE

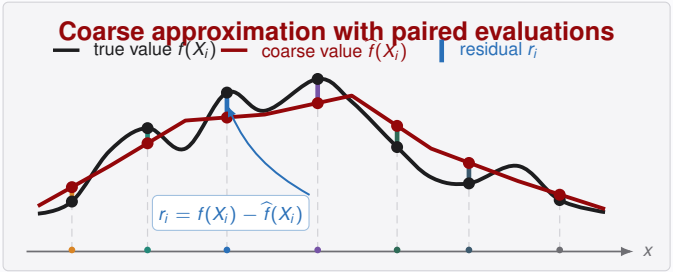
$$X_i \overset{\text{iid}}{\sim} \mu, \qquad \widehat{I}_N^{\text{MC}} = \frac{1}{N}\sum_{i=1}^{N} f(X_i).$$

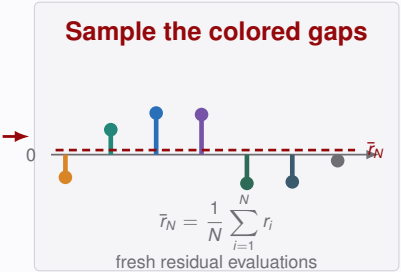
$$\mathbb{E}[\widehat{I}_N^{\text{MC}}] = I(f).$$

Sampling rate

$$\text{RMSE} = \frac{\sqrt{\text{Var}_\mu(f)}}{\sqrt{N}}.$$

Learn the bulk, simulate only the defect





$$I(f) = I(\widehat{f}) + \mathbb{E}[r(X)], \qquad \widehat{I}_{\mathrm{CV}} = I(\widehat{f}) + \bar{r}_N.$$

Training learns the bulk; inference averages only what the surrogate missed.

Residual Monte Carlo multiplies the two rates

Condition on a surrogate trained with independent data.

$$\mathbb{E}[\widehat{I}_{\mathrm{CV}} \mid \widehat{f}] = I(f), \qquad \mathrm{RMSE}(\widehat{I}_{\mathrm{CV}} \mid \widehat{f}) \le \frac{\|f - \widehat{f}\|_{L^2(\mu)}}{\sqrt{N}}.$$

Two independent gains multiply

$$\underbrace{\|f - \widehat{f}_m\|_2}_{O(m^{-\gamma})} \times \underbrace{N^{-1/2}}_{\text{residual samples}} = \underbrace{O(m^{-\gamma} N^{-1/2})}_{\text{corrected RMSE}}.$$

MATCHED BUDGETS: $N \asymp m$ $\qquad \mathrm{RMSE} = O(m^{-\gamma-1/2})$

Source: Blanchet, Chen, Lu, and Ying, arXiv:2305.16527, Theorems 4 and 5 for the noisy linear functional.

Moment estimation with rare peaks

Estimate a moment from random function values

$$\mathcal{D}_n = \{(X_i, f(X_i))\}_{i=1}^n, \quad X_i \overset{\text{iid}}{\sim} \text{Unif}(\Omega) \longrightarrow T_q(f) = \int_\Omega f(x)^q \, dx = \mathbb{E}[f(X)^q]$$

$\Omega = [0,1]^d$, n exact function values

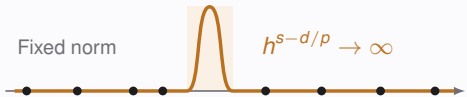
$q = 1$: integral $q \geq 2$: nonlinear moment

Sobolev regularity allows rare peaks

$$\mathcal{F} = \left\{ f : \|f\|_{W^{s,p}(\Omega)} \leq 1 \right\}, \quad \alpha = s/d$$

$$W^{s,p} \hookrightarrow L^{p^*}, \qquad \frac{1}{p^*} = \frac{1}{p} - \alpha$$

$0 < \alpha < 1/p$: finite p^*, no uniform height bound



Support volume $\asymp h^d$, $h \to 0$

What is the optimal estimation rate?

$$\mathcal{R}_n^\star = \inf_{\widehat{T} = \widehat{T}(\mathcal{D}_n)} \sup_{f \in \mathcal{F}} \mathbb{E} \left| \widehat{T} - T_q(f) \right|$$

Moment theorem: integer $q \geq 2$, $p > 2$, $q < p < 2q$. Expected absolute error. Boundary cases omitted.

Source: Blanchet, Chen, Lu, and Ying, arXiv:2305.16527, Section 2, Figure 1, Sections 3.1 and 3.2.

Optimal algorithms and minimax rates

Truncated Monte Carlo

$$\widehat{T}_{\mathrm{tr}} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{clip}_M(f(X_i))^q$$

$M \asymp n^{1/p-\alpha}$

Tail bias $\asymp$ sampling error

Regression control variate

$$\widehat{T}_{\mathrm{CV}} = \int_\Omega \widehat{f}^{\,q}\,dx + \frac{2}{n}\sum_{i=n/2+1}^{n}\left[f(X_i)^q - \widehat{f}(X_i)^q\right]$$

Fit $\widehat{f}$ on the first $n/2$ observations

Unbiased residual correction

$\alpha_0 = \frac{1}{p} - \frac{1}{2q}$ — Variance threshold: $p^* = 2q$

$\alpha_1 = \frac{2q-p}{p(2q-2)}$ — Rate crossover: spike = smooth

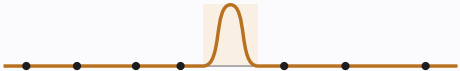
$\alpha = s/d$

Minimax error $\mathcal{R}_n^\star$

$\mathcal{R}_n^\star \asymp n^{q(1/p-\alpha)-1}$

$\mathcal{R}_n^\star \asymp n^{-\alpha-1/2}$

Spike



Unseen support, volume $\asymp n^{-1}$

Smooth function



Many small random perturbations

Expected absolute error. CV: optimal regression oracle, exact surrogate integral. Strict regimes, schematic axis.

Source: Blanchet, Chen, Lu, and Ying, arXiv:2305.16527, Theorems 1 to 3, Figure 2, Assumption 3.1.

Influence functions control nonlinear variance

Sample split estimation of $T_q(f) = \int_\Omega f^q$, $q \geq 2$

$$\widehat{T}_{q,\mathrm{CV}} = \int_\Omega \widehat{f}^{\,q} + \frac{2}{n} \sum\nolimits_{i=n/2+1}^{n} \left[f(X_i)^q - \widehat{f}(X_i)^q \right]$$

Influence expansion

$e = f - \widehat{f}$

$$f^q - \widehat{f}^{\,q} = \underbrace{q f^{q-1}}_{\mathrm{IF}_f(x)} e + R_q$$

$$\mathrm{Var}(\widehat{T}_{q,\mathrm{CV}} \mid \widehat{f}) \leq \frac{C_q}{n} \Bigg(\underbrace{\int_\Omega |f|^{2q-2} |e|^2}_{A\text{: influence fluctuation}} + \underbrace{\int_\Omega |e|^{2q}}_{B\text{: nonlinear propagation}} \Bigg)$$

Source: Blanchet, Chen, Lu, and Ying, arXiv:2305.16527, equation 7 and Theorem 2.

Proof mechanisms behind the phase diagram

$$e = f - \widehat{f}, \qquad f^q - \widehat{f}^{\,q} = \underbrace{q f^{q-1}}_{\mathrm{IF}_f} e + R_q, \qquad n\mathrm{Var}(\widehat{T}_{q,\mathrm{CV}} \mid \widehat{f}) \lesssim \underbrace{\|f^{q-1} e\|_2^2}_{A\text{: influence}} + \underbrace{\|e\|_{2q}^{2q}}_{B\text{: propagation}}.$$

$$\mathcal{R}_n^\star \asymp \max\left\{ n^{q(1/p-\alpha)-1}, n^{-\alpha-1/2} \right\}, \qquad \alpha = s/d, \quad p > 2, \quad q < p < 2q.$$

UPPER BOUNDS

Truncated MC	Regression CV	Regression CV	Regression CV
Tail bias $\asymp$ sampling error	Nonlinear propagation	Influence, dual norms	Bounded influence, $f \in L^\infty$
$M \asymp n^{1/p-\alpha}$	$\mathbb{E}[B] \lesssim n^{2q(1/p-\alpha)-1}$	$A \lesssim \|e\|_p^2$	$A \lesssim \|e\|_2^2$
$n^{q(1/p-\alpha)-1}$	$n^{q(1/p-\alpha)-1}$	$n^{-\alpha-1/2}$	$n^{-\alpha-1/2}$

$\alpha_0 = \frac{1}{p} - \frac{1}{2q}$ algorithm switch

$\alpha_1 = \frac{2q-p}{p(2q-2)}$ rate switch

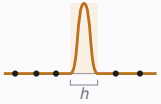
$\alpha_2 = \frac{1}{p}$ metric switch

$\alpha = s/d$

LOWER BOUNDS

$h^d \asymp n^{-1}$ K: smooth compact bump

Spike



$g(x) = h^{s-d/p} K\left(\frac{x-c}{h}\right)$

Unseen support, $\Pr(\mathrm{miss}) \asymp 1$

$\int g^q \asymp n^{q(1/p-\alpha)-1}$

Smooth function



random $\pm h^s$ bumps

$f_\eta(x) = M + h^s \sum_j \eta_j K\left(\frac{x-c_j}{h}\right)$

Sign bias $\asymp n^{-1/2}$, $\eta_j \in \{\pm 1\}$

$\Delta T_q \asymp h^s n^{-1/2} = n^{-\alpha-1/2}$

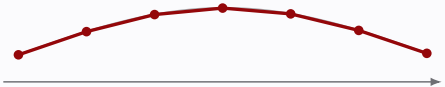
Expected absolute error, bounded Sobolev ball. Boundary cases omitted. CV upper bounds: regression oracle.

Source: Blanchet, Chen, Lu, and Ying, arXiv:2305.16527, Theorems 1 to 3, Figure 2, Appendix B.2.

A two scale method

$$f(x) = \underbrace{\widehat{f}(x)}_{\text{smooth, coarse scale}} + \underbrace{r(x)}_{\text{fine scale residual}} .$$

Smooth component



Function approximation

Small scale, high frequency residual



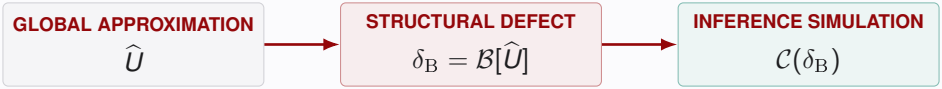
Monte Carlo

$$\widehat{I} = \int_\Omega \widehat{f}\, d\mu + \frac{1}{N}\sum_{i=1}^{N} r(X_i), \qquad X_i \overset{\text{iid}}{\sim} \mu.$$

$$\mathrm{RMSE}(\widehat{I} \mid \widehat{f}) = \frac{\sqrt{\mathrm{Var}_\mu(r)}}{\sqrt{N}} \le \frac{\|r\|_{L^2(\mu)}}{\sqrt{N}}.$$

Small residual energy controls the variance, not frequency alone.

Bellman residuals lift the same identity from integrals to paths



$$U_{\mathrm{corr}} = \widehat{U} + \mathcal{C}(\delta_{\mathrm{B}}).$$

INTEGRATION	SCIENTIFIC ML	GENERATIVE AI
$\widehat{f}$	$\widehat{u}$	$\widehat{V}$ or H
$f - \widehat{f}$	PDE Bellman residual	Bellman or Doob defect
residual Monte Carlo	Feynman Kac or MLP	SMC, Girsanov, or two queries

The defect is computable because the governing structure is known.

APPLICATION 1

Scientific machine learning

SCaSML: simulate the model's defect.

Linear case: subtraction gives the defect law

EQUATION 1 GOVERNING EQUATION: THE BELLMAN EQUATION

$$\partial_t u + \mathcal{L}u = f, \qquad u(T,\cdot) = g,$$

EQUATION 2 ANOTHER BELLMAN EQUATION THAT THE SURROGATE SATISFIED

$$\partial_t \widehat{u} + \mathcal{L}\widehat{u} = \widehat{f}, \qquad \widehat{u}(T,\cdot) = \widehat{g}.$$

EQUATION 1 MINUS EQUATION 2

$$\underbrace{(\partial_t + \mathcal{L})(u - \widehat{u})}_{(\partial_t+\mathcal{L})\breve{u}} = \underbrace{f - \widehat{f}}_{\text{Bellman error } \delta_{\mathrm{B}}}$$

Law of Defect: Value evaluation with Bellman Error

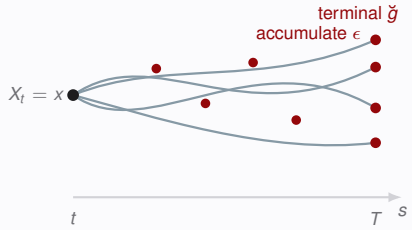
$$\breve{u} := u - \widehat{u}, \quad \breve{g} := g - \widehat{g}, \quad \delta_{\mathrm{B}} := f - \widehat{f} = -\epsilon,$$

$$\boxed{(\partial_t + \mathcal{L})\breve{u} = \delta_{\mathrm{B}}, \qquad \breve{u}(T,\cdot) = \breve{g}.}$$

Feynman Kac simulates only the defect

SAMPLE THE GENERATOR

$X_t = x, \qquad dX_s = \mu(s, X_s)\, ds + \sigma(s, X_s)\, dW_s.$



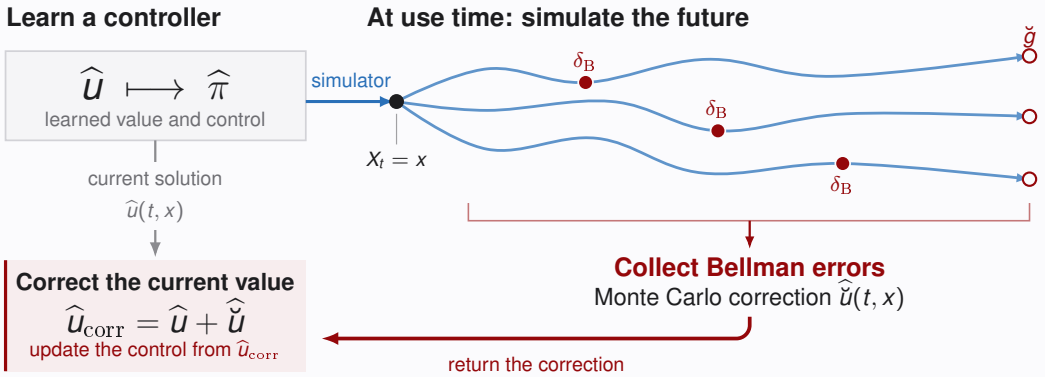
FEYNMAN KAC REPRESENTATION

$$\begin{aligned}\Delta &:= \breve{g}(X_T) - \int_t^T \delta_{\mathrm{B}}(s, X_s)\, ds \\ &= \breve{g}(X_T) + \int_t^T \epsilon(s, X_s)\, ds, \\ \breve{u}(t,x) &= \mathbb{E}_{t,x}[\Delta].\end{aligned}$$

MONTE CARLO REALIZATION

$$\widehat{u}^{(N)}_{\mathrm{corr}}(t,x) = \widehat{u}(t,x) + \frac{1}{N}\sum_{n=1}^{N} \Delta^{(n)}.$$

Simulator feedback for a learned controller



Source: SCaSML, Law of Defect and stochastic correction.

Defect size controls the Monte Carlo variance

MONTE CARLO FACTOR

$$\mathrm{Var}\left(\widehat{u}^{(N)}_{\mathrm{corr}} \mid \widehat{u}\right) = \frac{\mathrm{Var}(\Delta \mid \widehat{u})}{N}.$$

DEFECT FACTOR Condition on $\widehat{u}$ and assume $\|\breve{g}\|_\infty, \|\epsilon\|_\infty < \infty$.

$$|\Delta| \le \|\breve{g}\|_\infty + (T-t)\|\epsilon\|_\infty,$$

The errors multiply

$$\|\breve{g}\|_\infty \le C_T e(\widehat{u}), \|\epsilon\|_\infty \le C_\epsilon e(\widehat{u}) \quad \Longrightarrow \quad \boxed{\mathrm{RMSE}_{\widehat{u}} \le \left(C_T + (T-t)C_\epsilon\right)\frac{e(\widehat{u})}{\sqrt{N}}.}$$

Use the surrogate as a control variate.

Better surrogate model $\rightarrow$ Lower correction variance

Nonlinear case: subtraction gives the defect law

Semi-Linear Equation

$$\mathcal{N}[v] := \partial_t v + \mathcal{L}v + F\left(v, \sigma^\top \nabla v\right).$$

EQUATION 1

$$\mathcal{N}[u] = f, \qquad u(T) = g.$$

EQUATION 2

$$\mathcal{N}[\widehat{u}] = \widehat{f}, \qquad \widehat{u}(T) = \widehat{g}.$$

EQUATION 1 MINUS EQUATION 2

$$\partial_t \check{u} + \mathcal{L}\check{u} + F(\widehat{u} + \check{u}, \widehat{z} + \check{z}) - F(\widehat{u}, \widehat{z}) = \underbrace{f - \widehat{f}}_{\text{Bellman error } \delta_B} = -\epsilon,$$

$$\check{u}(T, \cdot) = g - \widehat{g}.$$

Law of Defect: $\partial_t \check{u} + \mathcal{L}\check{u} + \check{F}(\check{u}, \check{z}) = 0.$

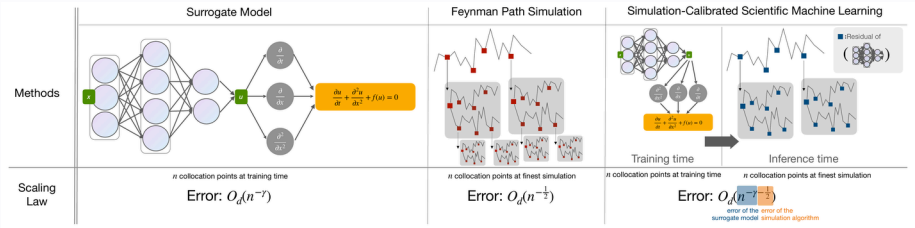
Key Observation: Closeness

$$\check{F}(v, z) := F(\widehat{u} + v, \widehat{z} + z) - F(\widehat{u}, \widehat{z}) + \epsilon$$

$\check{u} := u - \widehat{u}$, $\widehat{z} := \sigma^\top \nabla \widehat{u}$, $\check{z} := \sigma^\top \nabla \check{u}$.

Training and inference exponents add

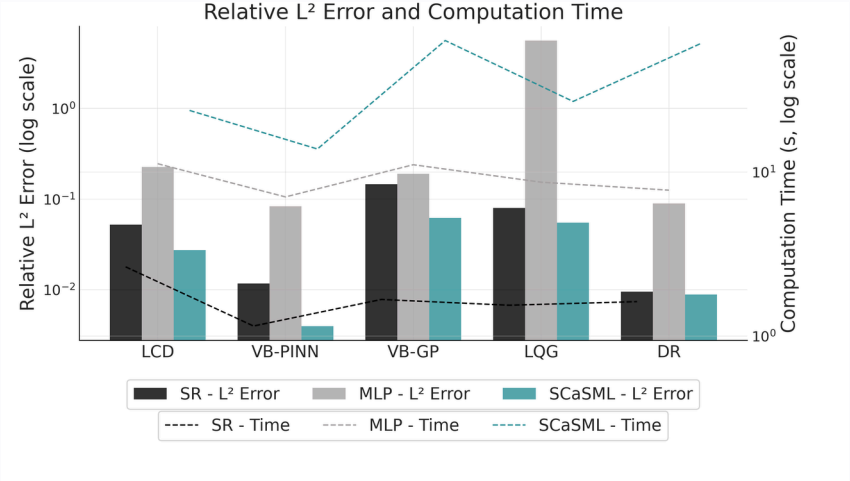
$$e(\widehat{u}) = O(m^{-\gamma}), \qquad N \asymp m \quad \Longrightarrow \quad \boxed{\mathrm{Err}_{\mathrm{SCaSML}} = O\Big(m^{-\gamma} N^{-1/2+o(1)}\Big) = O\Big(m^{-\gamma-1/2+o(1)}\Big).}$$



TRAINING γ $+$ **INFERENCE** $\frac{1}{2}$ $=$ **CORRECTED EXPONENT** $\gamma + \frac{1}{2}$

Monte Carlo multiplies the surrogate error; it does not replace it.

SCaSML spends inference compute to buy accuracy



More inference compute lowers relative L^2 error across all five benchmarks.

APPLICATION 2

Diffusion model

URGE: correct approximate guidance with path weights.

Guidance turns “at work” into a sampling direction

Suppose a diffusion model already knows how José looks.

BASE DISTRIBUTION $X_t \sim p_t$, public images of José



REWARD MODEL

Is José at work?

$G = \mathbb{P}(\text{José at work})$

extra drift follows predicted reward

$V(t)^2 \nabla G_t(X_t)$

GUIDED SAMPLE synthetic target



synthetic illustration • José at work

The base model supplies identity; the reward specifies the task; guidance changes the proposal.

Guidance follows the value, not the reward

BACKWARD VALUE

The terminal reward is propagated to time t through

$$h(x,t) = \mathbb{E}_{\mathbb{P}}[e^{\mathbf{r}(X_T)} \mid X_t = x],$$

Intractable in general

Ideal Doob guidance

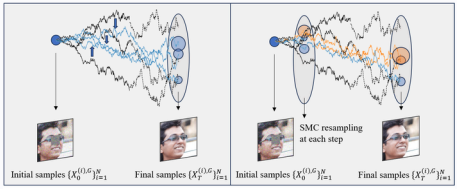
$$V(t)^2 \nabla \log h(x,t)$$

value gradient, **not** reward gradient $V(t)^2 \nabla_x r(x,t)$

PRACTICAL PROPOSAL

$$G \approx \log h \implies V(t)^2 \nabla \log h \longrightarrow V(t)^2 \nabla G.$$

$G \neq \log h \implies \mathbb{P}^G \neq \mathbb{Q}$: **correction is required.**



(a) Naïve Guidance (b) Unbiased Guidance via SMC

Reward scores the endpoint.

Value guides the path.

Girsanov gives the exact path correction

GUIDED PROPOSAL TO BASE DYNAMICS

$$\frac{d\mathbb{P}}{d\mathbb{P}^G}(X^G_{0:t}) = \exp\left(-\int_0^t V\nabla G^\top dW - \frac{1}{2}\int_0^t V^2\|\nabla G\|_2^2 ds\right).$$

REWARD TILT

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(X_{0:t}) \propto e^{r(X_t,t)-r(X_0,0)}.$$

COMBINED PATH CORRECTION

$$\boxed{\frac{d\mathbb{Q}}{d\mathbb{P}^G}(X^G_{0:t}) \propto \exp\left[r(X^G_t,t) - r(X^G_0,0) - \int_0^t V\nabla G^\top dW - \frac{1}{2}\int_0^t V^2\|\nabla G\|_2^2 ds\right].}$$

Key advantage

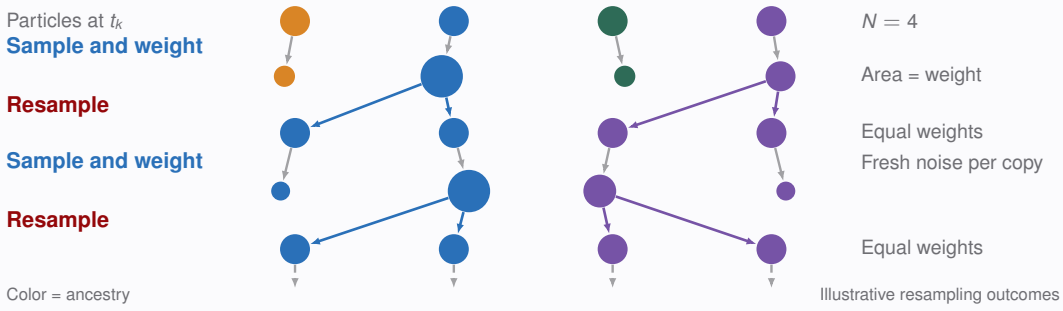
The correction weight needs no reward gradient, reward Laplacian, or score.

URGE corrects diffusion paths sequentially

$$X_{k+1}^{i,G} = X_k^{i,G} + (v_k^i + V_k^2 \nabla G_k^i)\Delta t_k + V_k \sqrt{\Delta t_k}\, \xi_k^i, \qquad \xi_k^i \sim \mathcal{N}(0, I).$$

$$\log \beta_k^i = r(X_{k+1}^{i,G}, t_{k+1}) - r(X_k^{i,G}, t_k) - V_k (\nabla G_k^i)^\top \sqrt{\Delta t_k}\, \xi_k^i - \frac{1}{2} V_k^2 \|\nabla G_k^i\|_2^2 \Delta t_k.$$

Sample particles → **Update path weights** → **Normalize and resample**



URGE is exact in an iterated limit

Iterated consistency

Under the paper's regularity assumptions and correct initial law,

$$\lim_{N\to\infty}\lim_{\Delta t\to 0}\mathbb{E}_{\bar{\mathbb{P}}}[M_T^{N,\Delta t}(\varphi)] = \frac{\mathbb{E}_{\mathbb{P}}[e^{\mathbf{r}(X_T)-r(X_0,0)}\varphi(X_T,T)]}{\mathbb{E}_{\mathbb{P}}[e^{\mathbf{r}(X_T)-r(X_0,0)}]}.$$

Scope of the theorem

Asymptotic consistency only: the result gives no finite N total variation or sample complexity bound.

Path weights recover the particle correction rate

Define the terminal conditioned instantaneous path weight rate

$$\lambda(x,t) = \lim_{s \uparrow t} \frac{\mathbb{E}_{\mathbb{P}^G}[w_{s,t}^{\mathrm{URGE}} \mid X_t^G = x] - 1}{t - s}.$$

If the guided marginal density is sufficiently smooth, then $\boxed{\lambda(x,t) = w_{\mathrm{particle}}(x,t).}$

Previous Works:

$$w_{\mathrm{particle}}(x,t) = \partial_t r - \frac{V^2}{2}(\Delta r + \|\nabla r\|^2) + \nabla r^\top (v + V^2 \nabla G) + V^2 \nabla_x \log p^\top \nabla_x (G - r) + V^2 \Delta G$$

PATH SPACE

A weight for the simulated path segment

$w_{s,t}^{\mathrm{URGE}}$

Path weights; URGE still resamples particles.

$\lambda = w_{\mathrm{particle}}$ only as $s \uparrow t$

$\not\equiv$

finite Δt, N

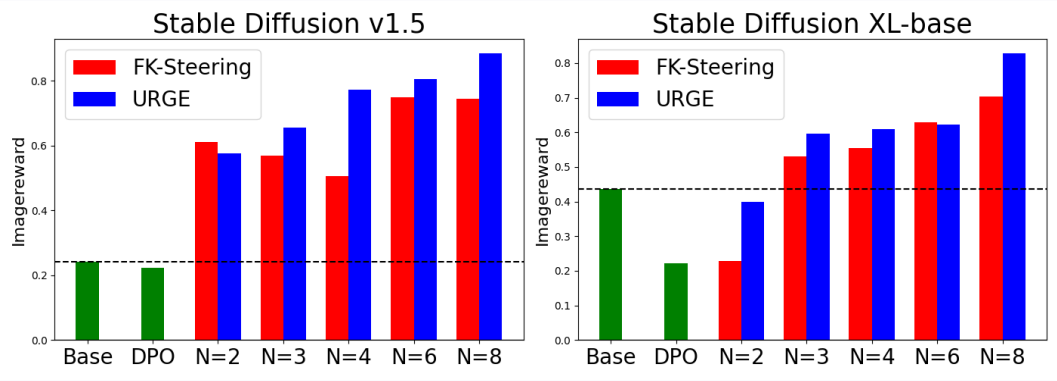
PARTICLE RESAMPLING

Local selection at each solver step

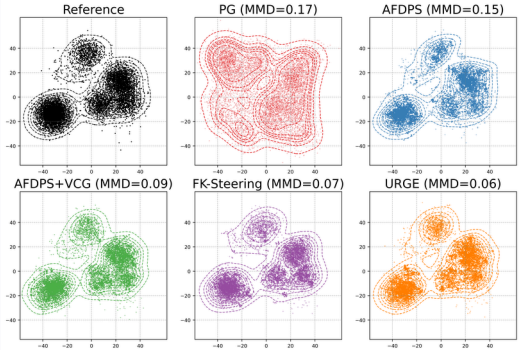
$w_{\mathrm{particle}}(x,t)$

Kills or duplicates particles.

Inference scaling with more particles



Pathwise correction recovers the target modes



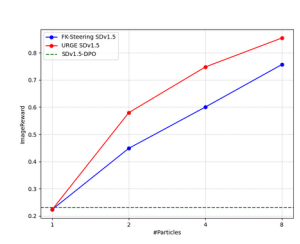
On the reported 30 dimensional Gaussian mixture, URGE gives the closest sample cloud to the analytical target, with **MMD** 0.06.

From inverse problems to neural rewards

FFHQ 256 SUPER RESOLUTION

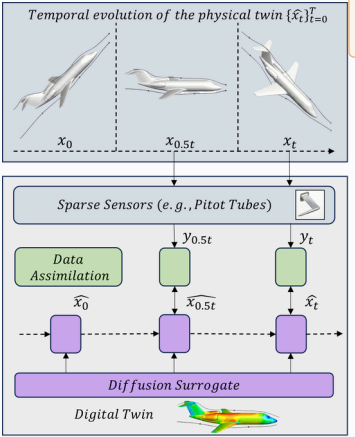


SD V1.5 PARTICLE SCALING



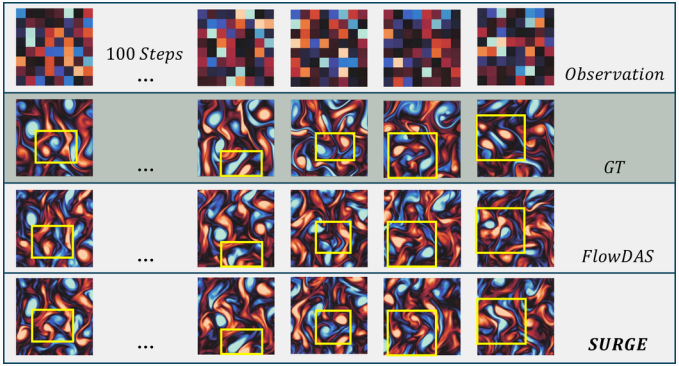
URGE matches derivative based restoration; ImageReward exceeds FK Steering for $N = 2, 4, 8$.

Data Assimilation



REWARD

Reward measures alignment with reality.

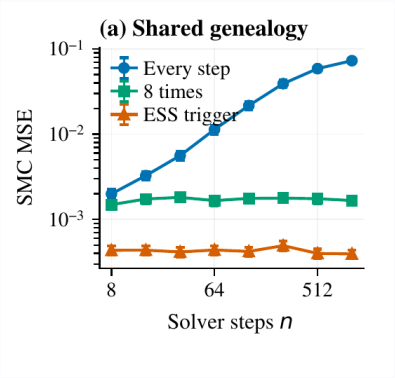


APPLICATION 3

From SMC to Stein guidance

SMC numerics reveal shared ancestry; sequential rejection removes copying; value-only correction removes the remaining bias.

SMC numerics under mesh refinement



ONLY SMC VARIANTS ARE COMPARED Same target and value approximation; only the numerical mesh and resampling schedule change.

What the plot says

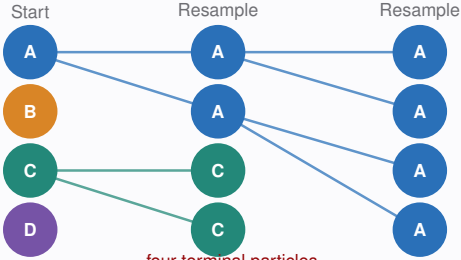
- Resampling at every solver step makes the MSE grow rapidly with depth n.
- A fixed number of resampling times or an ESS trigger remains nearly flat.

Instability tracks population interaction, not merely the local diffusion step.

Source: Supplied numerical experiment, `fig_stein_experiments.pdf`, panel (a).

Shared ancestry explained the SMC error

REPEATED RESAMPLING COPIES ENTIRE PREFIXES



four terminal particles
one surviving stochastic family

WHY LOCAL DIAGNOSTICS MISS IT

Everything can look healthy

Flat weights can give ESS $= N$; even every one-particle marginal can be correct.

But the empirical mean sees pair covariance

For $\bar{f}_N = N^{-1} \sum_i f(X_n^i)$,

$$\mathrm{Var}(\bar{f}_N) = \frac{1}{N} \mathrm{Var}(f(X_n^1)) + \frac{N-1}{N} \mathrm{Cov}(f(X_n^1), f(X_n^2)).$$

***N* endpoints do not mean *N* independent histories.**

Source: Supplied manuscript, neutral Brownian SMC construction and variance decomposition.

Perfect ESS can hide nonconvergence

Full multinomial resampling at every microstep; $X_0^{i,N} = 0$, $n, N \geq 2$.

NEUTRAL BROWNIAN CHAIN

$$X_p^{i,N} = X_{p-1}^{A_p^i,N} + n^{-1/2}\xi_p^i,$$

$$A_p^i \sim \mathrm{Unif}\{1,\dots,N\}, \qquad \xi_p^i \sim \mathcal{N}(0,1).$$

Every local check passes

$\phi \equiv 1$, $\mathrm{ESS}_p = N$, and
$X_n^{i,N} \sim \mathcal{N}(0,1)$ exactly.
Every one particle marginal is exact.

Yet the empirical mean does not stabilize

For $g(x) = \sin(x)/2$,

$$\mathbb{E}\left[\left(\frac{1}{N}\sum_{i=1}^{N} g(X_n^{i,N})\right)^2\right] \geq \frac{1}{32\mathrm{e}^2}\min\left\{\frac{n}{N}, 1\right\}.$$

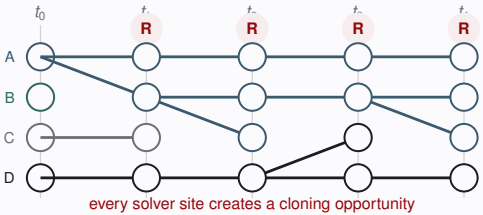
For fixed N, the error stays nonzero as $n \to \infty$.

$\mathrm{RMSE} \leq \varepsilon$ **forces** $N = \Omega(n\varepsilon^{-2})$ **and work** $= \Omega(n^2\varepsilon^{-2})$**.**

Source: User supplied manuscript, *Two Queries, No Gradients*, Theorem 1 and Corollary 2; counterexample restored from the original talk.

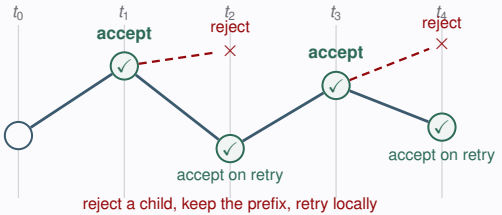
Sequential rejection retries children, not paths

FULL MICRO RESAMPLING one parent pool



categorical ancestor draw from one shared particle pool

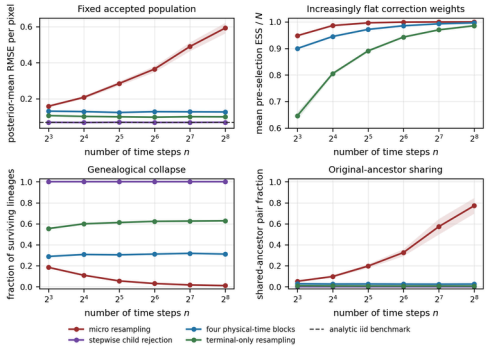
EXACT CHILD REJECTION one proposal stream



one accepted trajectory with local candidate rejections

Resampling moves mass across lineages. Rejection spends serial trials within a lineage.

Rejection stays stable as SMC genealogy collapses



CONTROLLED LINEAR IMAGING

$N = 160$ accepted paths · 36 runs

$n = 8, 16, 32, 64, 128, 256$

Sequential rejection

$$\mathrm{RMSE} : 0.0683 \text{ to } 0.0696$$

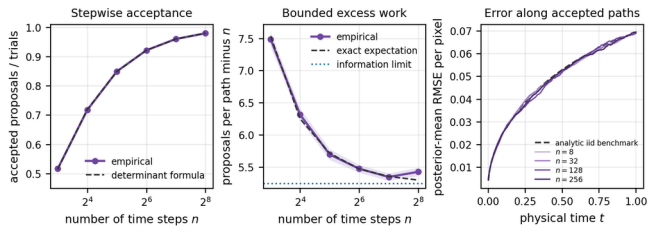
Micro-resampling SMC

$$\mathrm{RMSE} : 0.159 \text{ to } 0.593$$

At the finest mesh, ESS is $0.99994N$, yet only 1.02% of lineages survive and 77.2% of terminal pairs share an ancestor.

Rejection stays on the iid benchmark while repeated SMC cloning collapses genealogy.

Rejection trades shared ancestry for local retries



$$W_n = \sum_{k=1}^{n} \alpha_{k,n}^{-1} = n + I(X_1; Y) + O(n^{-1})$$

Acceptance: $0.517 \to 0.979$

Excess proposals: $7.54 \to 5.30$

Information limit: $I(X_1; Y) = 5.239$ nats

Information controls retry overhead; population controls independent sampling variance.

Source: Supplied diffusion imaging manuscript, rejection diagnostics and exact Gaussian work formula.

Rejection loses SMC's running correction weight

SMC CARRIES PROBABILITY COMPENSATION

Particle state

$$\left(X_{0:k}^i, W_k^i\right) \qquad W_{k+1}^i = W_k^i \, r_k^i$$

The incremental factor r_k^i records how far the approximate value function is from Bellman consistency along that path. **Path and correction weight travel together.**

PURE REJECTION RETURNS AN UNWEIGHTED PATH

The local normalizer is hidden

$$G_k(x) = (P_k H_{k+1})(x),$$

$$Q_k^H(x, dy) = \frac{P_k(x, dy) H_{k+1}(y)}{G_k(x)}.$$

The accept/retry decision samples Q_k^H, but it does not retain G_k as a path weight. Approximate local tilts therefore need not compose to the target.

The missing whole-path compensation

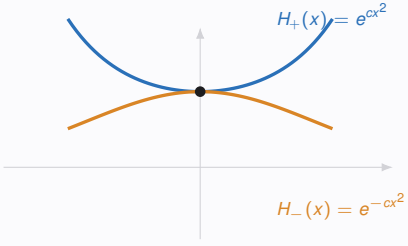
$$R_H = H_0(x_0) \prod_{k=0}^{n-1} \frac{G_k(X_k)}{H_k(X_k)}$$

Generate first; correct once at the end.

Source: Supplied manuscript and Lu (2026), path-law factorization for approximate Doob values.

A direct final correction asks for derivatives

SAME VALUE AND GRADIENT, DIFFERENT AVERAGE



$$H_\pm(0) = 1, \quad H'_\pm(0) = 0, \; c > 0$$

$$\mathbb{E}H_\pm(\sqrt{h}Z) = 1 \pm ch + O(h^2)$$

GUIDANCE USES A FIRST DERIVATIVE

$$dX_t = \left(b_t + a_t \nabla \log H_t\right) dt + \sigma_t \, dB_t.$$

The direct residual route

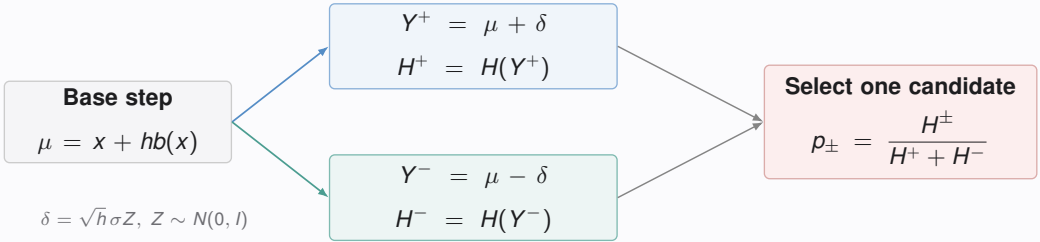
$$\mathcal{E}_t = \frac{\partial_t H_t + b_t \cdot \nabla H_t + \frac{1}{2} a_t : \nabla^2 H_t}{H_t}.$$

$$R_H = H_0(x_0) \exp\left(\int_0^T \mathcal{E}_t(X_t) \, dt\right).$$

Goal: use function values only—no ∇H and no $\nabla^2 H$.

Source: Lu (2026), blog, Sec. 5; Zhu et al. (2026), Sec. 3 and 4. Continuous identity assumes a valid change of measure.

Two symmetric queries choose a direction



Value contrast controls selection

$$\mathbb{E}[Y^J - \mu \mid Z] = \delta \frac{H^+ - H^-}{H^+ + H^-}.$$

The even part gives an average

$$S = \frac{H^+ + H^-}{2}.$$

Keep S for the later path correction.

Source: Lu (2026), blog, Sec. 6; Zhu et al. (2026), Sec. 3.1.1, Eq. (5).

Why the direction is a log gradient

SUBTRACT: RECOVER A DIRECTIONAL SLOPE

Odd part

$$H^+ - H^- \approx 2\delta^\top \nabla H(\mu).$$

ADD: NORMALIZE BY THE VALUE

Even part

$$H^+ + H^- \approx 2H(\mu).$$

SELECTION COMBINES BOTH OPERATIONS

$$\frac{H^+ - H^-}{H^+ + H^-} \approx \delta^\top \nabla \log H(\mu).$$

Then average over the random direction:

$$\mathbb{E}[\delta\delta^\top] = h\sigma\sigma^\top.$$

Two value queries recover the leading Doob drift

$$\mathbb{E}[Y^J \mid x] = \mu + h\sigma\sigma^\top \nabla \log H(\mu) + o(h).$$

This is a small-step mean-drift statement for positive smooth H, not exact finite-step sampling.

Source: Zhu et al. (2026), Sec. 3.1.1, Eq. (5); Lu (2026), blog, Sec. 6.

Stein links value selection to gradient guidance

GAUSSIAN INTEGRATION BY PARTS

Stein identity

$$\mathbb{E}[Zf(Z)] = \mathbb{E}[\nabla f(Z)], \quad Z \sim N(0, I).$$

Set $Y = \mu + \sqrt{h}\sigma Z$, $f(Z) = H(Y)$:

$$\mathbb{E}[ZH(Y)] = \sqrt{h}\,\sigma^\top \mathbb{E}[\nabla H(Y)].$$

Random direction $\times$ value

reveals an average slope.

EXACT LOCAL REJECTION, WHEN AN ENVELOPE EXISTS

Propose Y from the base Gaussian. Accept with probability $H(Y)/M$, where $0 < H \le M$.

$$G_h(\mu) = \mathbb{E}H(\mu + \sqrt{h}\sigma Z).$$

The accepted mean

$$\mathbb{E}[Y \mid \text{accept}] = \mu + h\sigma\sigma^\top \nabla_\mu \log G_h(\mu).$$

The exact identity uses the smoothed value G_h. It matches a mean, not a shifted Gaussian law.

Finite pools replace these expectations with parallel value queries.

Source: Lu (2026), blog, Sec. 7; Zhu et al. (2026), Eq. (6), Eq. (23), and Proposition 6.

A centered pool extends two queries to any K

A BALANCED SET OF RANDOM DIRECTIONS

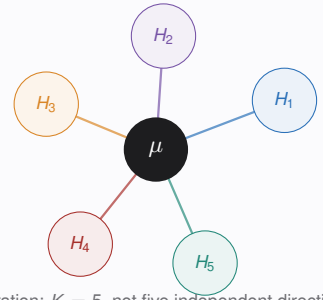


Illustration: $K = 5$, not five independent directions

Each $Z_i \sim N(0, I)$; pool directions are dependent.

$K = 2$: $Z_1 = -Z_2$. **Odd K also works.**

DRAW, CENTER, AND RESTORE MARGINAL VARIANCE

$$G_i \overset{\text{iid}}{\sim} N(0, I), \qquad \bar{G} = \frac{1}{K}\sum_i G_i,$$

$$Z_i = \sqrt{\frac{K}{K-1}}(G_i - \bar{G}), \qquad \boxed{\sum_i Z_i = 0}.$$

Query in parallel, then select

$$Y_i = \mu + \sqrt{h}\,\sigma Z_i, \qquad H_i = H(Y_i),$$

$$\Pr(J = i \mid U) = \frac{H_i}{\sum_j H_j}, \qquad S = \frac{1}{K}\sum_i H_i.$$

Source: Zhu et al. (2026), Sec. 3.1.2, Eqs. (7) and (8); Lu (2026), blog, Sec. 7.

Centering cancels the leading pool noise

THE SAME POOL AVERAGE WILL ENTER THE PATH WEIGHT

$$S = \frac{1}{K}\sum_i H(\mu + \sqrt{h}\sigma Z_i) = H(\mu) + \underbrace{\sqrt{h}\, g^\top \bar{Z}}_{\text{linear random term}} + O_{L^2}(h), \qquad g = \sigma^\top \nabla H(\mu).$$

Independent candidates

$\mathbb{E}\bar{Z} = 0$, but $\bar{Z}$ varies from pool to pool.

$\sqrt{h}$ scale fluctuations

The linear term vanishes only after averaging.

Centered candidates

$\bar{Z} = 0$ for every pool.

h scale fluctuations

The linear term cancels before averaging.

A better arrangement of queries, not an extra derivative evaluation.

Small-step comparison with fixed K, positive smooth values, and controlled second-order remainder. The smoothing bias is a separate issue.

Source: Zhu et al. (2026), Remark 2; Lu (2026), blog, Sec. 7.

The same queries provide the path weight

AT EACH SOLVER STEP

One pool, two outputs

$$S_k = \frac{1}{K}\sum_{i=1}^{K} H_{k+1}(Y_{k,i})$$

Choose X_{k+1} with probability proportional to its value.

Keep both S_k and $H_{k+1}(X_{k+1})$.

No extra value query. No value derivative.

ACCUMULATE THE PROBABILITY COMPENSATION

Weight for a complete proposal

$$R = \phi(X_n)\prod_{k=0}^{n-1}\frac{S_k}{H_{k+1}(X_{k+1})}.$$

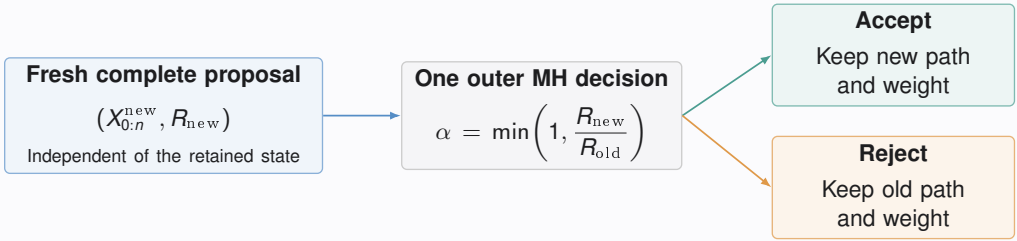
$$\log R = \log \phi(X_n) + \sum_{k=0}^{n-1} \log \frac{S_k}{H_{k+1}(X_{k+1})}.$$

Valid on the augmented space of pools and choices, not by an arbitrary density plug-in.

Relative values guide. The same pool average corrects.

Source: Lu (2026), blog, Sec. 8; Zhu et al. (2026), Algorithm 1 and Theorem 12.

Whole-path MH corrects after generation



What is independent?

Complete proposals can be generated independently and batched in parallel.

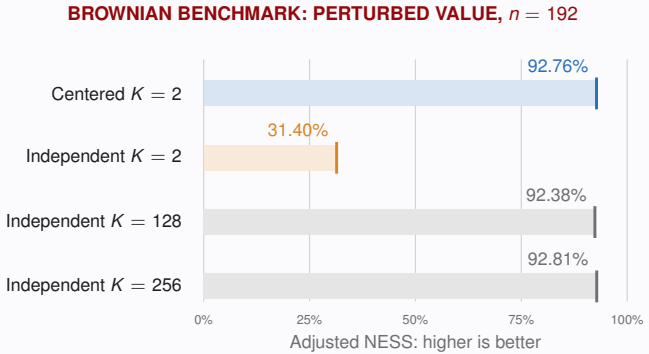
What is not independent?

Consecutive outputs of one MH chain may repeat and remain correlated.

The target is invariant; a finite MH run can still have initialization error.

Source: Lu (2026), blog, Sec. 8; Zhu et al. (2026), Algorithm 2.

Two centered candidates stabilize weights



Two queries suffice in this test

92.76%

Independent pools require 128 or 256 candidates to reach a similar level in this test.

Adjusted NESS evaluates normalizer estimation. It is not image quality or the number of independent MH outputs.

Source: Zhu et al. (2026), Sec. 5.1 and Figure 2; Lu (2026), blog, Sec. 10. Values redrawn from the reported results.

Whole-path correction improves the target match

Hopper: fixed diffusion policy and critic, averaged over eight fixed states.

CONDITIONAL CRITIC-SCORE KS DISTANCE: LOWER IS BETTER

Target mass	Guided only	With path MH
30%	0.2174	**0.0306**
60%	0.2804	**0.0697**
80%	0.4041	**0.1128**

Target mass is placed on the original policy's top 10% of actions by critic score.

Same two-candidate guidance

$K = 2$ at every diffusion step.

Correction uses 32 complete proposals:
1 initialization + 31 outer updates.

What the experiment measures

Critic-score distribution matching, not realized return.

A useful direction does not remove the need for probability correction.

Source: Zhu et al. (2026), Sec. 5.2, Table 2, Appendix H.3; Lu (2026), blog, Sec. 10.

Approximate rewards guide reasoning prefixes

Youheng Zhu and Yiping Lu, ICML 2026

FIXED LLM, TERMINAL OBJECTIVE

X_t is the reasoning prefix after t steps.
The base LLM supplies the next step kernel P_t.

$$\pi_\phi(x_{0:T}) \propto p_{\mathrm{ref}}(x_{0:T})\,\phi(x_T).$$

A prefix value looks ahead to the final reward:

$$V_t^*(x) = \mathbb{E}_{\mathrm{ref}}[\phi(X_T) \mid X_t = x], \qquad V_T^* = \phi.$$

THE SAME BELLMAN DEFECT LAW

For an approximate value $\widehat{V}_T = \phi$, define

$$\mathcal{B}_t := \widehat{V}_t - P_t\widehat{V}_{t+1}, \qquad e_t := V_t^* - \widehat{V}_t.$$

$$\boxed{e_t = P_t e_{t+1} - \mathcal{B}_t, \qquad e_T = 0.}$$

The mismatch between a prefix and its expected continuations measures local value error.

SMC allocates compute to promising prefixes

Generate from the base LLM, weight by $G_t = \widehat{V}_t(X_t)/\widehat{V}_{t-1}(X_{t-1})$, and resample. The value ratios telescope to the terminal reward.

Source: Zhu and Lu, ICML 2026, Sections 2–3.

Bellman error controls the cost of reasoning

Assume positive values and the adjacent value ratio bound $L^{-1} \le \widehat{V}_{t+1}(x_{t+1})/\widehat{V}_t(x_t) \le L$.

UNIFORM MULTIPLICATIVE BELLMAN ERROR

$$\frac{1}{1+\varepsilon} \le \frac{P_t \widehat{V}_{t+1}(x)}{\widehat{V}_t(x)} \le 1+\varepsilon.$$

RATIO CONTROL ALONE

Worst case cost for constant TV accuracy:

$$\mathrm{TC} = \Omega\left(L^{2T/3}\right)$$

For $L > 1$, with queried output paths.

SMC WITH APPROXIMATE VALUES

A sufficient particle count for TV error δ is

$$N \ge \frac{L^6 T (1+\varepsilon)^{6(T-1)}}{2\delta}.$$

A SUFFICIENT REGIME FOR POLYNOMIAL INFERENCE COST

$$\varepsilon = O(1/T) \quad \Longrightarrow \quad \mathrm{TC}_{\mathrm{SMC}} = O(NT) = O\left(\frac{L^6 T^2}{\delta}\right).$$

TV error is for the averaged output law. T counts reasoning steps. Unit cost per sample and value query.

Source: Zhu and Lu, ICML 2026, Assumptions 3.1–3.2, Theorems 4.1 and 5.1, Corollary 5.2.

One correction principle, four numerical languages

Setting	Learned object	Structural defect	Inference correction
Integration	$\widehat{f}$	$f - \widehat{f}$	residual Monte Carlo
Scientific ML	$\widehat{u}$	PDE or Bellman residual	Feynman Kac or multilevel Picard
Diffusion	G or H	Doob or path law defect	Girsanov weights or Stein pools with path MH
LLM	Prefix value $\widehat{V}_t$	$\widehat{V}_t - P_t \widehat{V}_{t+1}$	SMC reweighting and resampling

Inference time scaling law

$$U_{\mathrm{corr}} = \underbrace{\widehat{U}}_{\text{learned bulk}} + \underbrace{\mathcal{C}(\delta_{\mathrm{B}})}_{\text{simulate the structural miss}} .$$

Learn the bulk once; spend inference compute only on the miss.

Thank you